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# Pioneering Carbon Steps Indonesia: Implementation CO2 Injection of Mature Carbonate Reservoir in East Java Field

D. Halinda, O. Muraza, A. Sundja, I. Maulana, M. Arifin, M. Nasfiah, L. Akhmadi, and I. Rosyadi, Pertamina, Jakarta, Indonesia; A. B. Prasetyo, SKK Migas, Jakarta, Indonesia; H. Okabe and K. Mikami, Japan Organization for Metals & Energy Security, Tokyo, Japan; K. Kato, Japan Petroleum Exploration Company, Tokyo, Japan



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## Abstract

$CO_2$ enhanced oil recovery ($CO_2$ EOR) is a proven technology that has been widely applied to many oil fields, especially in North America. Recently, interest in $CO_2$ EOR has been growing in the context of carbon capture, utilization and storage (CCUS), as it can reduce greenhouse gas emissions while supporting the current energy foundation (Núñez-López and Moskal 2019). Several pilot tests have been conducted outside of North America recently, including in Abu Dhabi (Al Hajeri et al. 2010), Japan (Koga et al. 2024), Kuwait (Pradhan et al. 2024), and Oman (Nasralla, Gavnholt, et al. 2025).

In Southeast Asia, only a few pilot tests using the huff-and-puff method have been conducted with a high-purity $CO_2$ injection (Uchiyama et al. 2012; Halinda et al. 2023), and commercial $CO_2$ EOR projects have not yet been reported. One challenge we recognize is that reservoir pressures tend to be below the minimum miscibility pressures (MMP) of $CO_2$ due to the high geothermal gradients in the region, which are typically 4-6°C/100 m (Zhu et al. 2024) (Figure 1). Therefore, understanding the recovery efficiency of $CO_2$ EOR under immiscible / near-miscible conditions is crucial to realizing the potential of $CO_2$ EOR in this region.

## Introduction

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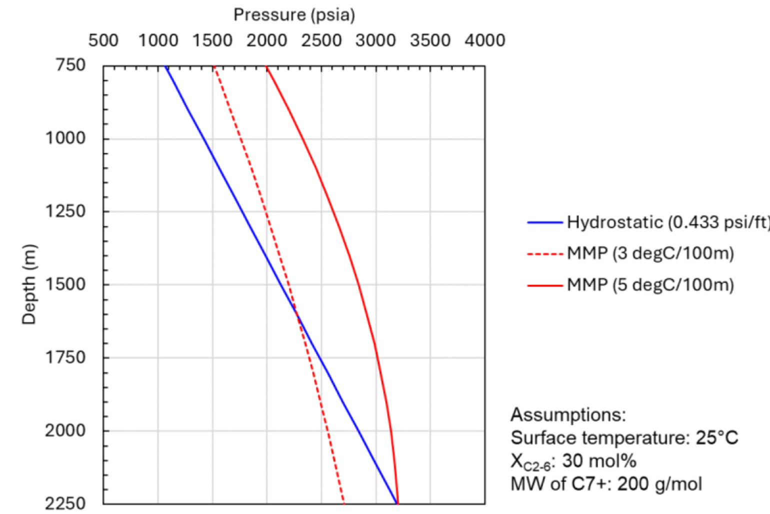


**Figure 1—Minimum miscibility pressure (MMP) for geothermal gradients of 3°C/100 m and 5°C/100 m. The MMP was calculated using the correlation by Yuan et al. (2004). When a high geothermal gradient is applied, the estimated MMP can be much higher than hydrostatic pressure.**

## Field Description

The field under study is the Sukowati field, a mature onshore oil field located in East Java, Indonesia. The Jambaran-Tiung Biru (JTB) gas field is a potential $CO_2$ source for the future full-field $CO_2$ EOR application. The JTB gas field contains more than 20 mol% $CO_2$ in its produced gas and is located approximately 35 km from the Sukowati field. Due to the favorable source-sink conditions for CCUS, the application of $CO_2$ EOR in this field has been extensively studied (Marbun et al. 2021; Panggabean et al. 2022).

The main reservoir of the Sukowati field consists of a carbonate reef characterized with a remarkably high anticline structure, reaching over 250 meters in height from the initial oil-water contact (Figure 2). Production in the field began in 2004 with natural depletion, followed by the reinjection of produced water in 2015. There are 39 wells, including four (4) water injectors. Throughout its history, the perforation intervals of the producers have frequently been shifted upward by workover operations to produce the remaining oil in the upper layers. The water-cut of the field is currently about 90%, indicating that the reservoir is nearly watered out, leaving behind bypassed oil. Since the main recovery mechanism is the bottom water drive, our understanding of reservoir heterogeneity and inter-well connectivity is limited.

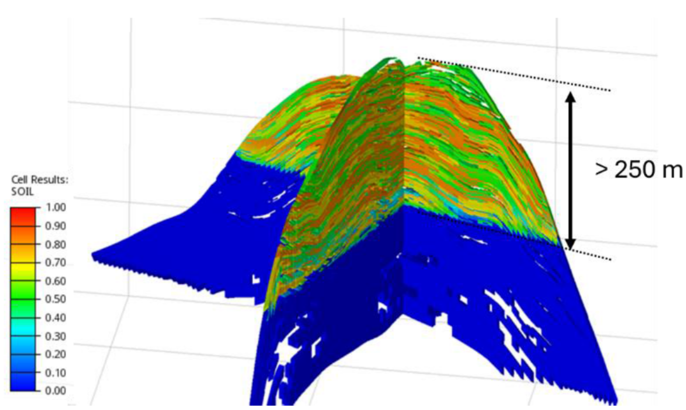


**Figure 2—3D view of the Sukowati reservoir simulation model. The color map shows the initial oil saturation.**

The reservoir pressure has been maintained due to strong pressure support from the bottom aquifer. The current reservoir pressure is approximately 2,600 psia, which is about 300 psi below the initial reservoir pressure. The region has a high geothermal gradient; the reservoir temperature is approximately 130°C, which is equivalent to a geothermal gradient of over 5°C/100 m. The original reservoir oil was nearly saturated at the initial reservoir pressure and contains more than 20 mol% of $CO_2$. Due to the high reservoir temperature and fluid composition, the equation of state (EOS) calibrated with the laboratory experiments predicts a high minimum miscibility pressure (MMP) of approximately 3,100-3,300 psia for pure $CO_2$, even though the crude oil is light with an API gravity of 38-40°. Although the MMP estimation remains uncertain due to the uncertainty of the fluid composition and the EOS parameters, miscible flooding is unlikely to be achieved under the given conditions of the field. It is therefore important to understand the recovery efficiency of $CO_2$ injection under near-miscible conditions. The field test was primarily designed for this purpose.

## Design and Operations of Inter-Well $CO_2$ Injection Test

The $CO_2$ injection test was designed using the existing wells in the field. The target well for the $CO_2$ injection was carefully selected based on the locations, perforated intervals and production performance of the surrounding producers. The selected injector, SKW-26, is a shut-in well located at the crestal area of the carbonate reef, surrounded by three active producers, namely SKW-02, SKW-05 and SKW-22ST, which are the monitoring producers in this test (Figure 3). SKW-05 has the shortest distance from the injector (150 m), followed by SKW-22ST (180 m) and SKW-02 (250 m).

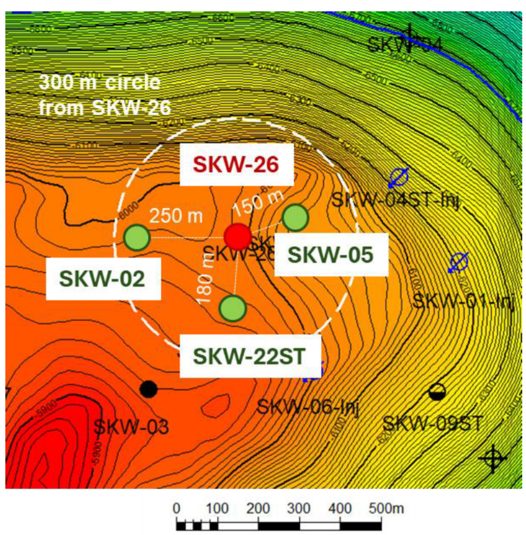


**Figure 3—Depth structure map of the reservoir top around the test area. The selected $CO_2$ injector, SKW-26, is surrounded by three active producers, SKW-02, SKW-05 and SKW-22ST.**

In this test, high-purity $CO_2$ (> 99%) was continuously transported in liquid form from the industrial gas sources via truck. The $CO_2$ injection volume is one of the most critical design parameters because it must be sufficient to achieve the test purpose, but it can also significantly impact on the project costs. Hence, the perforation interval of the injector was optimized to ensure that the production response could be observed with a smaller $CO_2$ injection volume (Figure 4). Since the perforation interval of SKW-22ST may be in a different geological unit than the other wells, limited connectivity with the other wells was expected. The target volume of the $CO_2$ injection was 2,500 tons (~50 MMscf), with a target injection rate of 100 tons/day (~2 MMscf/day), constrained by the $CO_2$ logistics rather than the well injectivity.

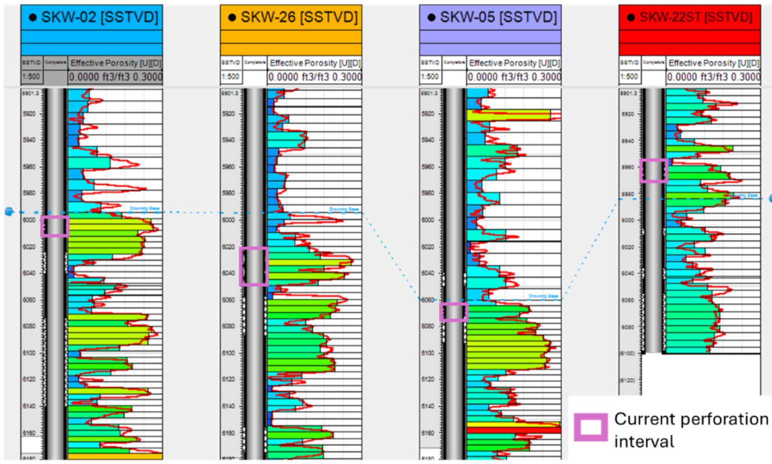


**Figure 4—Well correlation of the test area showing porosity and perforation intervals. The perforation intervals of SKW-02, SKW-05 and SKW-26 are well correlated, while that of SKW-22ST may be located in a different geological unit.**

The liquid $CO_2$ transported via truck was pressurized by a liquid pump and its temperature was controlled by an electric indirect heater installed after the pump (Figure 5). The injection equipment is a mobile unit with advanced data gathering and real-time monitoring system. The details of the $CO_2$ injection operation are described in Halinda et al. (2025).

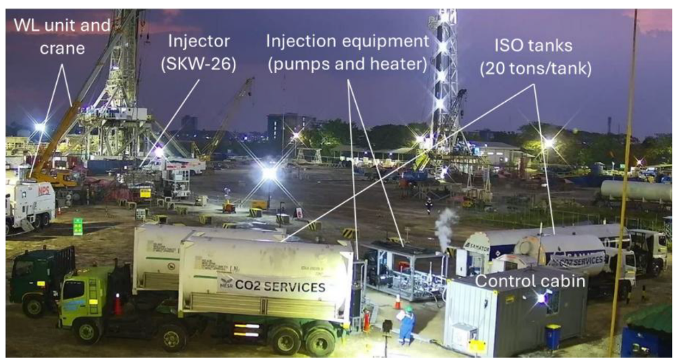


**Figure 5—Overview of the injection site. The $CO_2$ tanks transported by truck were directly connected to the mobile injection skids, including a liquid pump and electric heater.**

A monitoring plan was prepared for both the injector and the producers. Since there were no permanent bottomhole gauges available for the injector, real-time monitoring of the bottomhole pressure (BHP) and temperature (BHT) was implemented via wireline to ensure the safe injection operations. Additionally, there was concern that the injected $CO_2$ could not be clearly identified by the monitoring of produced gas compositions due to the high natural $CO_2$ content in the reservoir fluid. Therefore, a passive gas tracer, a perfluorocarbon tracer of 10 kg, was injected on the first day of the $CO_2$ injection.

For the producers, a dedicated three-phase test separator was installed to accurately measure the flow rates. Gas and liquid samples were taken from the test separator at least once a day and analyzed by an on-site laboratory. The monitoring period was for three months, determined by a pre-injection simulation study, as in addition of the tracer monitoring for the tracing of the perfluorocarbon been done in the producer well through gas sampling and been analyzed with multi-component GC.

# Results of Inter-Well $CO_2$ Injection Test

## $CO_2$ Injection Performance

The $CO_2$ injection was completed as planned, with no critical issues. A total of 2,600 tons of $CO_2$ was injected for a period of one month. Figure 6 shows the injection profiles during the test. Although some operational interruptions caused short shut-ins, a stable injection rate of approximately 80-100 tonnes/day was achieved. The BHP and BHT data show some fluctuations caused by periodic running-in and -out hole operations of the wireline tools, which were necessary to maintain the internal temperature of the tools in an acceptable range due to the high reservoir temperature and extended time in the wellbore.

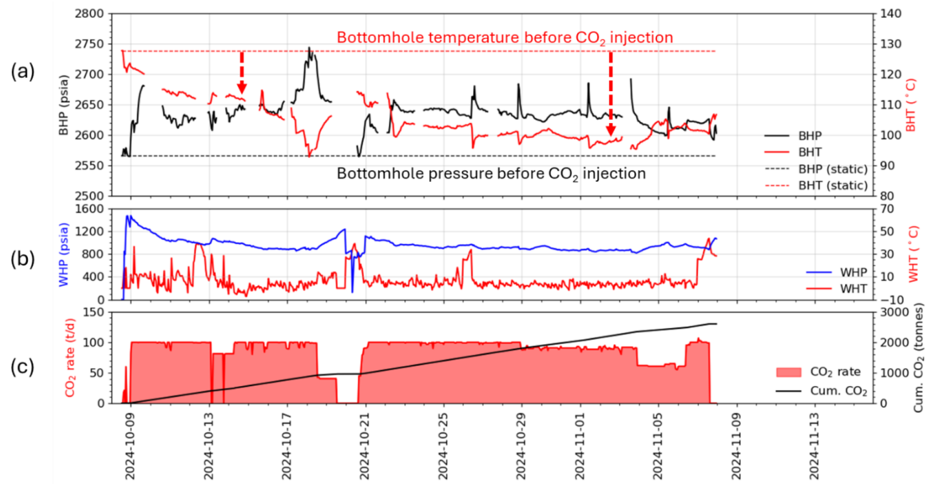


**Figure 6—$CO_2$ injection profile of SKW-26. (a) Bottomhole pressure and temperature; (b) Wellhead pressure and temperature; (c) $CO_2$ injection rate and cumulative injected mass volume.**

The wellhead temperature (WHT) of the injected $CO_2$ was maintained around 0-5°C by the heater. Due to the cold $CO_2$ injection, the BHT decreased from 128°C to approximately 100°C. The wellhead pressure (WHP) also decreased due to the temperature drop in the wellbore, which increased the $CO_2$ density and hydrostatic gradient. The WHP stabilized at around 900 psia, indicating that the $CO_2$ was in the liquid phase at the wellhead. Furthermore, the well exhibits good injectivity of approximately 1.4 tonnes/day/psi with a perforation interval of only nine (9) meters (Figure 7). No clear relationship was observed between the $CO_2$ injectivity and BHT in this test.

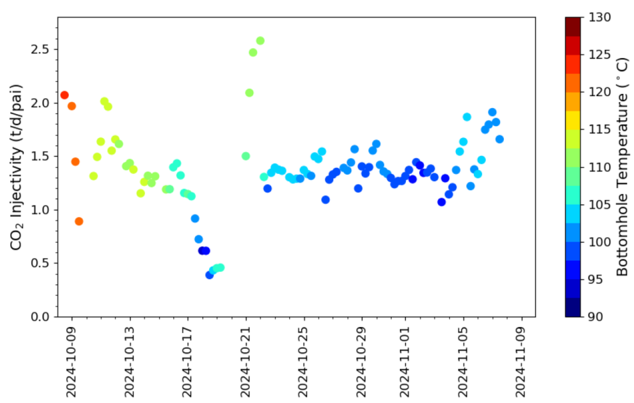


**Figure 7—$CO_2$ injectivity of SKW-26 color-coded by bottomhole temperature. No clear relationship was observed between $CO_2$ injectivity and bottomhole temperature.**

## Production Performance

Figure 8 shows the production performance of the three (3) monitoring producers. After 22 days from the start of the $CO_2$ injection, SKW-02 clearly showed an increase in oil production rate and a decrease in water-cut. This positive trend continued for at least two (2) months, until the end of December 2024. Conversely, the gas production rate did not clearly respond to the $CO_2$ injection. SKW-05 was shut in for a workover and reopened two (2) weeks before the start of the $CO_2$ injection. Since the baseline performance of this well remains highly uncertain, no significant incremental oil was counted in this study based on the performance before and after the workover. This well also did not respond clearly to the $CO_2$ injection in terms of the gas production rate. SKW-22ST experienced a slight increase in oil production rate in early December 2024 due to a sudden increase in liquid production rate. However, this increase was not considered to be due to the $CO_2$ injection, since the well has experienced the similar performance changes in the past, and the water cut did not decrease. This well also did not clearly respond to the $CO_2$ injection in terms of the gas production rate. In summary, a positive and clear response to the $CO_2$ injection was observed in SKW-02, while no clear responses were observed in the other wells, and there were no indications of measurable $CO_2$ breakthrough in any of the wells. As of the end of December 2024, the cumulative incremental oil volume had exceeded 5,000 bbl, equivalent to a $CO_2$ utilization factor of less than 10 Mscf/bbl.

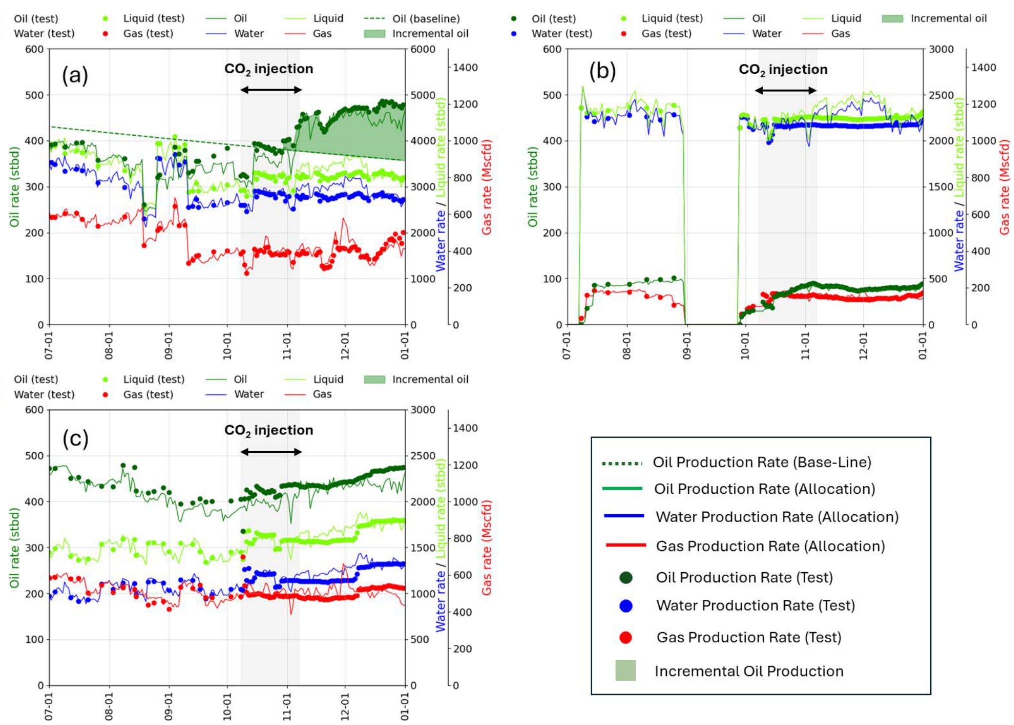


**Figure 8—Production profiles of the three (3) monitoring producers. (a) SKW-02; (b) SKW-05; (c) SKW-22ST. A positive and clear response to the $CO_2$ injection was observed in SKW-02, while no clear responses were observed in the other wells.**

Figure 9(a) shows the gas tracer concentrations of the separator gas samples from the three (3) monitoring producers. While there was no clear increase in gas rate indicating $CO_2$ breakthrough observed in the production data, the tracer data did show clear breakthrough of the gas tracer after 28 days at SKW-05, 54 days at SKW-02, and 80 days at SKW-22ST. This data is valuable because it confirms that all the monitoring producers are communicating with the $CO_2$ injector, and the timing of the breakthroughs indicates the inter-well connectivity in the area.

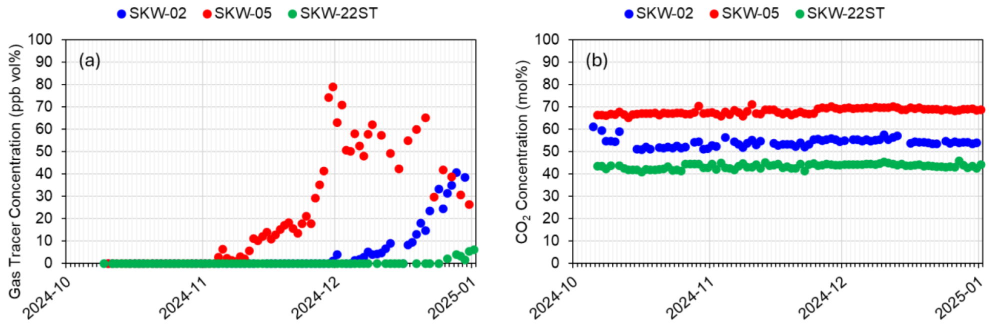


**Figure 9—Analysis results of separator gas samples. (a) Gas tracer concentration; (b) $CO_2$ concentration. Tracer breakthrough was observed after 28 days at SKW-05, 54 days at SKW-02, and 80 days at SKW-22ST, while there was no clear indication from the $CO_2$ concentration data.**

Figure 9(b) shows the $CO_2$ concentrations of the separator gas samples. This data does not clearly indicate $CO_2$ breakthrough. Additionally, baseline $CO_2$ concentrations vary significantly by well. This variation may be due to the presence of in-situ $CO_2$ dissolved in the produced water in this field. Most of the dissolved gas in produced water is $CO_2$ because it is highly soluble in water. If only water were produced, the $CO_2$ concentration would be nearly 100% pure. Taking this effect into account, the $CO_2$ concentration in produced gas can be expressed in terms of the gas-water production ratio:

$$\begin{aligned} x_{CO_2} &= x_{CO_2}^{brine}\left(\frac{R_{sb}Q_w}{Q_g}\right) + x_{CO_2}^{HC}\left(\frac{Q_g - R_{sb}Q_w}{Q_g}\right) \\ &= x_{CO_2}^{brine}\left(\frac{R_{sb}}{GWR}\right) + x_{CO_2}^{HC}\left(1 - \frac{R_{sb}}{GWR}\right), \end{aligned} \tag{1}$$

where $x_{CO_2}$ is the $CO_2$ concentration in produced gas [mol%], $x_{CO_2}^{brine}$ is the $CO_2$ concentration in reservoir brine [mol%], $R_{sb}$ is the solution gas-water ratio of reservoir brine [scf/stb], $x_{CO_2}^{HC}$ is the $CO_2$ concentration in hydrocarbon gas [mol%], $Q_g$ is the gas production rate [scf] and $Q_w$ is the water production rate [stb], $GWR(=Q_g/Q_w)$ is the gas-water production ratio [scf/stb]. Figure 10 shows the extensive data obtained from the field, and the theoretical calculation by Eq.1. The theorical calculation assumes $x_{CO_2}^{brine} = 1$ and $x_{CO_2}^{HC} = 0.38$ based on the PVT data of the field. $R_{sb}$ was determined to match the field data. The actual field data aligned with the theoretical calculation by Eq.1. Therefore, the observed variation in $CO_2$ concentration by well is due to $CO_2$ solubility in water.

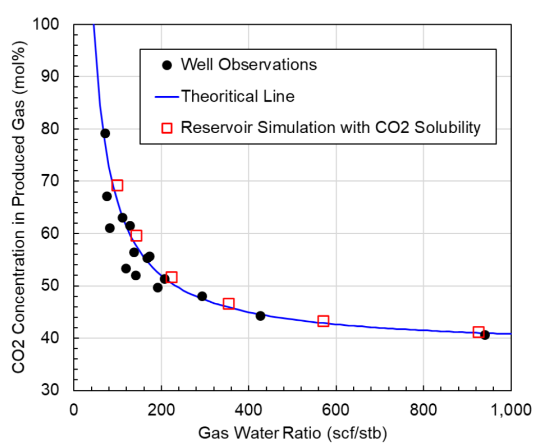


**Figure 10—$CO_2$ concretation in produced gas vs. gas water ratio. The actual field data aligned with the theoretical calculation by Eq.1 as well as reservoir simulation incorporating $CO_2$ solubility.**

## Reservoir Simulation Study

The field test successfully demonstrated that the $CO_2$ injection increased oil production efficiently in this reservoir. However, several questions remain, such as:

1. Why was the $CO_2$ breakthrough not clearly observed in any one of the monitoring producers, even though the tracer was?
2. What is the main mechanism of the increased oil production?
3. How can the results be upscaled?

Answering these questions requires validating and interpreting the test results using reservoir simulation. We developed a static sector model of the test area and conducted history matching of the field test using the commercial compositional reservoir simulator SLB Eclipse E300. This section describes our modeling approach and the results of the simulation.

### Sector modeling

***Grid, Boundary Conditions and Initialization.*** The same grid resolution (30 m by 30 m) with the full field model was used in the sector model. Local grid refinement (10 m by 10 m) was applied to the selected model for detailed history matching. Boundary conditions were applied to the model using pore volume multipliers to mimic the pressure support, especially from the bottom aquifer. The model was divided into four (4) regions, one region for each well in the area, and each region was assigned a constant water saturation (Figure 11). Equilibrium or constant pressure was applied to the entire model to match the actual static pressure observed during the test. Due to the limited size and duration of the test, the simplified boundary and initial conditions are acceptable for this study.

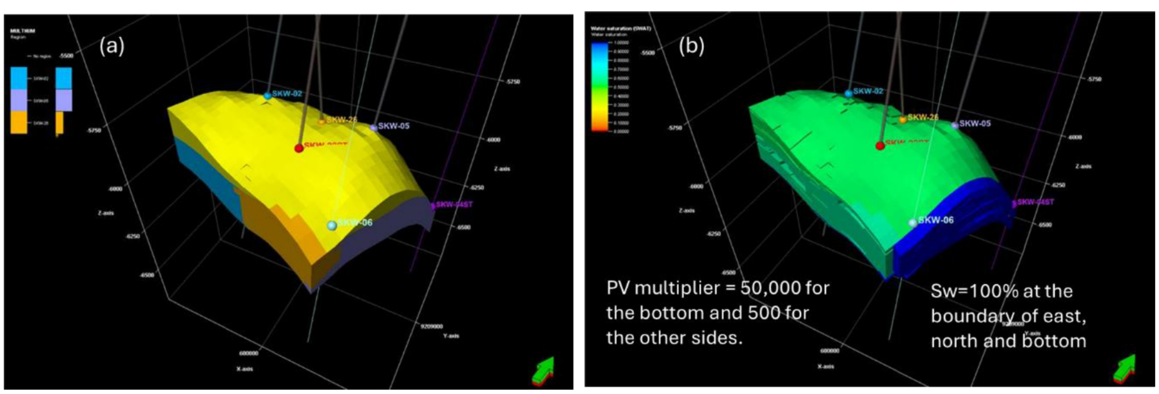


**Figure 11—Setting up the sector model (a) Definition of well regions; (b) Initial water saturation and boundary conditions.**

***Model Screening.*** To better capture the connectivity between wells, multiple realizations were generated from the static sector model by changing the distributions of facies and properties using different seed numbers. Figure 12 shows the results of these realizations in terms of gas production rate. Even in this small area, the inter-well connectivity varies significantly by realization. For example, some realizations show significant gas breakthrough to SKW-05, which was not observed during the actual test. Then, the model that best represented the inter-well connectivity was selected for further history matching.

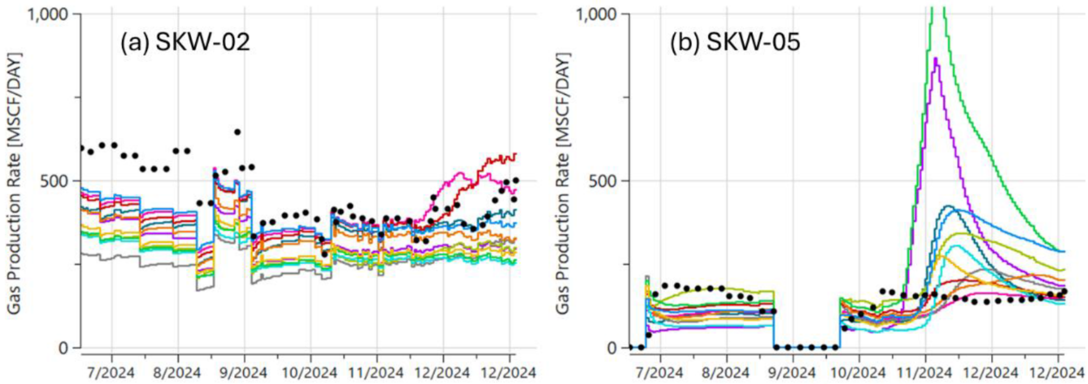


**Figure 12—Simulated gas production rates of multi-realizations. (a) SKW-02; (b) SKW-05. The markers show the production history, and the lines show the simulation results.**

***$CO_2$ Solubility in Water.*** For some high water-cut wells, including SKW-05, It was found that nearly half of the produced gas originates from the produced water. Therefore, $CO_2$ solubility was incorporated into the simulation model. The equilibrium calculation between water and hydrocarbons was modeled by using a commercial simulator function (Chang, Coats, and Nolen 1998), while the $CO_2$ solubility in water was explicitly calculated using the accurate solubility model (Duan and Sun 2003). The square symbols in Figure 10 show the results of the single-cell reservoir simulation that incorporated the $CO_2$ solubility. These results are consistent with the field data and the theorical calculation.

## History Matching of the Selected Model

The variables selected for the first step of the history match are, 1) the end-point of gas relative permeability ($k_{rg}$), and 2) the inter-region transmissibility. As expected from the slower gas breakthrough than anticipated observed during the test, it was necessary to reduce the end-point of gas relative permeability to the one-third of the original value. Masalmeh, Al-Keebali, and Igogo (2021) observed a similar low gas relative permeability in a carbonate reservoir at high water saturation. Further investigation is necessary, but lower gas relative permeability is a high possibility to explain both the slow gas breakthrough and the inter-well connectivity observed by the tracer data. The inter-region transmissibility was also calibrated to better replicate the timing of the tracer breakthrough. Using the calibrated model, two key variables controlling the incremental oil production were investigated, as described in the following subsections.

***Interfacial-Tension Dependent Oil-Gas Relative Permeability.*** Under near-miscible conditions, where oil-gas interfacial tension (IFT) values approaching to zero, displacement efficiency can be significantly affected by the change in relative permeability with varying IFT. Nasralla, et al. (2025) demonstrated that neglecting IFT-dependent relative permeability effects could underestimate the recovery factor and may reduce the benefits of near-miscible flooding. In this study, the IFT-dependent relative permeability was originally not considered due to the lack of validation data. However, in order to match the actual performance, it was necessary to incorporate these effects. In the simulator used, IFT dependent relative permeability is modeled by Coats (1980) using the interpolation function $F_k$ defined as:

$$F_k = \min\left[1, \left(\frac{\sigma}{\sigma_0}\right)^N\right], \tag{2}$$

where $\sigma_0$ is the reference IFT [dyne/cm] and N is the exponent. The oil-gas relative permeability is calculated by a linear interpolation between immiscible and miscible values based on $F_k$. In this study, the reference IFT was set at 2.5 dynes/cm and $N$ was changed from 0.25 (default) to 0.5.

Figure 13 shows the interpolation parameter used in this study and its influence on the simulation results for SKW-02. The IFT-dependent oil-gas relative permeability significantly impacts on the incremental oil production, and actual performance is better captured by incorporating this effect. This suggests that the injected $CO_2$ efficiently mobilized the remaining oil in this reservoir. Further validation, including laboratory experiments, may be necessary to confirm this effect and determine the appropriate parameters.

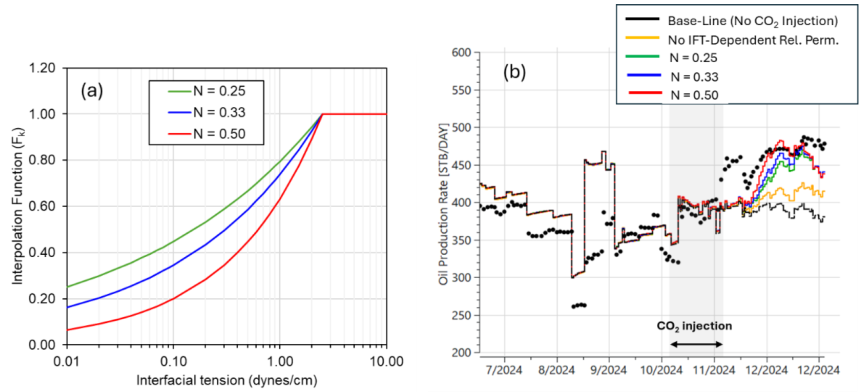


**Figure 13—IFT-dependent oil-gas relative permeability. (a) Interpolation functions used in this study; (b) Sensitivity of the interpolation functions to the oil production rate of SKW-02. The marker shows the production history, and the lines show the simulation results.**

***Miscibility.*** The EOS model used in this study has a MMP of 3,110 psia, which is 550 psi higher than the current reservoir pressure of 2,560 psia at the test depth. Although the EOS model was calibrated using the existing laboratory data, the MMP estimation remains uncertain due to the uncertainty of the fluid compositions and EOS parameters. Therefore, an alternative EOS model with a MMP of 2,640 psia was prepared, while still replicating the most of the existing laboratory data. Figure 14 compares the SKW-02 simulation results of the two EOS models. In both cases, $N = 0.25$ was used for the IFT-dependent relative oil-gas permeability. As expected, the EOS with the lower MMP can reduce the IFT further and increase oil production, even though both cases remain below the MMP. As it can be seen in the figure, to simulate the observed incremental oil rate given the assumptions for the IFT-dependent relative permeability, a lower MMP than the original EOS was required. One hypothesis is that the lower temperature of the injected $CO_2$ than the reservoir temperature (see Figure 6) decreased the MMP within the displacement zone. Therefore, it is important to validate the MMP and the IFT-dependent oil-gas relative permeability in a future study, which will be one of the main objective of the upcoming revisit of the dynamic modelling update of the Sukowati field.

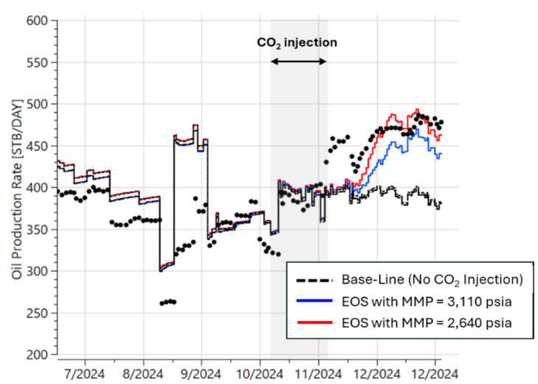


**Figure 14—Sensitivity of MMP to the oil production rate of SKW-02. The marker shows the production history, and the lines show the simulation results.**

***Results of History Matching.*** Figure 15 shows the results of the history matching using the EOS with the MMP of 2,640 psia. The production performance of each monitoring well was reasonably matched. Figure 16 shows the $CO_2$ concentration in the produced gas and the timing of the tracer breakthrough. The variation in $CO_2$ concentration was reasonably matched by incorporating the $CO_2$ solubility. Due to very limited breakthrough of injected $CO_2$, there are almost no changes in the $CO_2$ concentration in both simulation and observed data. Furthermore, the timing of the tracer breakthrough was also reasonably matched, indicating that the model accurately represents the inter-well connectivity. These results validate the reliability of the simulation model.

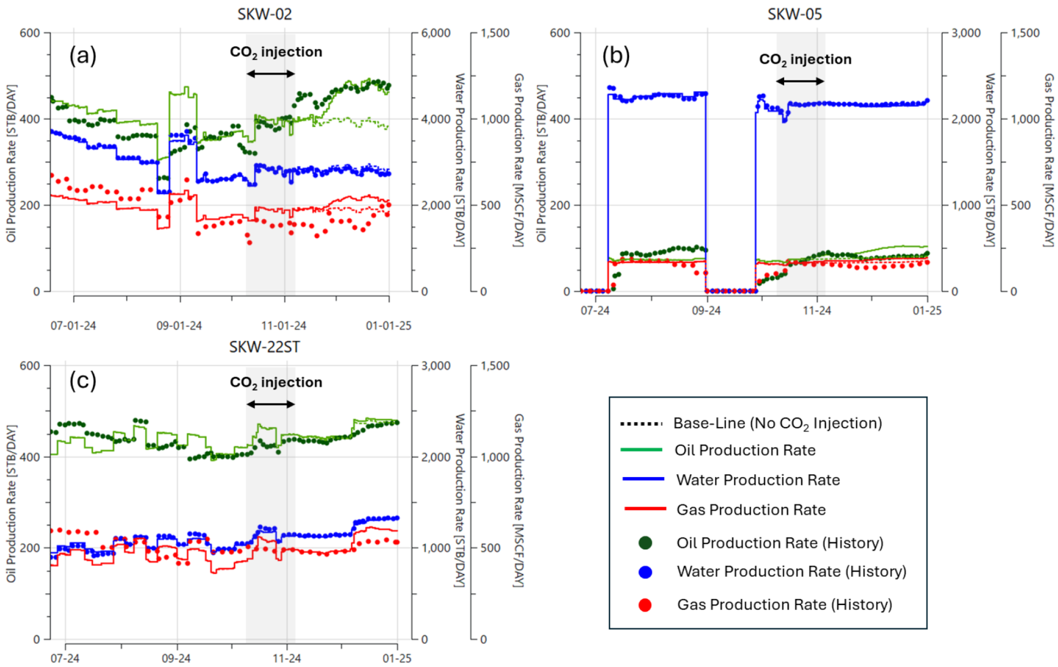


**Figure 15—Results of the history matching of the production performance. (a) SKW-02; (b) SKW-05; (c) SKW-22ST. The reasonable match was achieved for all three (3) monitoring producers.**

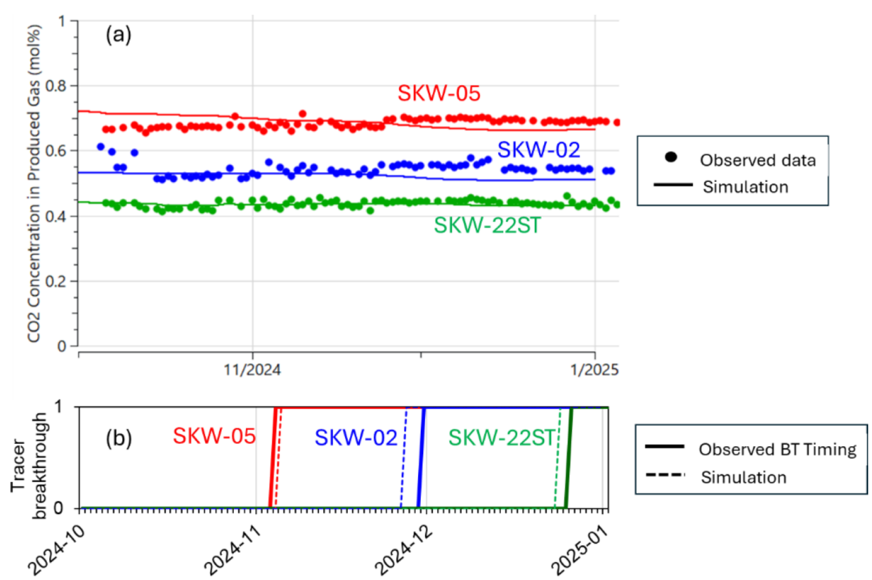


**Figure 16—Results of the history matching of the $CO_2$ concentration and tracer breakthrough timing. (a) $CO_2$ concentration in produced gas; (b)Timing of gas tracer breakthrough (0: No breakthrough, 1: Breakthrough).**

## Conclusions

This paper presents the results of an inter-well $CO_2$ injection test in a carbonate reservoir. The primary objective of this test was to validate the recovery efficiency of $CO_2$ injection under near miscible conditions, which is a key to unlock the potential of $CO_2$ EOR in Southeast Asia. This pilot serves as a critical step toward de-risking the full-field development of the Sukowati field and similar reservoirs. The following conclusions can be drawn from this study:

1. The $CO_2$ inter-well injection test successfully demonstrated efficient incremental oil recovery through $CO_2$ injection with a $CO_2$ utilization factor of less than 10 Mscf/stb, with the injection condition of $CO_2$ below the MMP.
2. The simulation study suggests that the injected $CO_2$ efficiently mobilized the remaining oil through the IFT-dependent oil-gas relative permeability effects under near miscible conditions and/or a lower MMP reservoir condition than expected.
3. Gas tracer data was highly valuable for identifying inter-well connectivity, especially in a reservoir with in-situ $CO_2$. Additionally, the amount of dissolved $CO_2$ in produced water may affect production performance, depending on the in-situ $CO_2$ content.

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